Nicolai Uhre Mandrup

Software Engineer · MSc Software Engineering (Expected 2026) · Copenhagen, Denmark

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PROFILE

Full-stack software engineer with a strong backend and data engineering focus, currently completing an MSc in Software Engineering at Aalborg University. I have delivered production-grade Python systems, full-stack web applications, ML pipelines, and BI solutions across both academic and professional settings. I bring hands-on experience in agile teams, a deep commitment to code quality, and the versatility to move between data science, backend architecture, security tooling, and business intelligence.

TECHNICAL SKILLS

Languages	Python · JavaScript · TypeScript · SQL · Java · VBA
Frameworks	React · React Native · Flask · Spring Boot · Node.js · Pydantic · Pandas · NumPy · Scikit-learn
Infrastructure	Docker · Kubernetes · Nginx · GitHub Actions · CI/CD · PostgreSQL · SQLite
BI & Data	Power BI · Excel VBA · Jupyter Notebooks · Matplotlib · Seaborn · Statistical Modelling
Practices	TDD · Property-based testing · mypy · Ruff · Swagger/OpenAPI · Agile / Scrum · REST APIs

WORK EXPERIENCE

Junior IT Consultant | IDA Insurance *2024 – 2025*

Supported data-driven decision-making across member administration and evaluation workflows at a major Danish professional association.

- Built and maintained **Power BI dashboards** to communicate member analytics and evaluation outcomes to stakeholders
- Automated reporting pipelines using **Excel VBA** and Microsoft 365, reducing manual effort and improving accuracy
- Led part of a **platform migration**, evaluating system architecture and optimising the new database structure
- Delivered ad hoc IT maintenance and digital tool improvements across member administration systems

Substitute & Kindergarten Teacher | Marie Kruses Skole, Farum *2020*

- Developed strong organisational and prioritisation skills, managing dynamic classroom environments

Volunteer | Kenya Wildlife Service, Nairobi *2020 (6 weeks)*

- Thrived in a fast-paced international environment, meeting project deadlines under pressure

Waiter | Halifax Burgers, Vesterbro *2018 – 2023*

Customer Service Representative | Call Me, Amager *2018 – 2019*

- Handled a broad range of technical and non-technical customer inquiries, building strong communication skills

SELECTED PROJECTS

Project	Stack	Highlights
QuantForge	Python · Pydantic · NumPy · pytest · GitHub Actions	Production-grade algo-trading backtester. 89 tests, property-based testing, cross-platform CI/CD, full analytics suite (Sharpe, VaR, CVaR).

Text-to-SQL Interface	JavaScript · Node.js · SQLite (WASM) · OpenAI API	Browser-native NL-to-SQL engine with in-memory SQLite, OpenAI integration, query explain plans, CSV export, and persistent history.
Network Vulnerability Scanner	Python · JavaScript · HTML/CSS	LAN device discovery, TCP port scanning, banner grabbing and heuristic risk reporting. Web UI with Markdown + JSON-output.
Hospital Navigation System	Java · Spring Boot · PostgreSQL · Docker · A*	Multi-tier backend with A* pathfinding for indoor routing. RESTful APIs, CI/CD automation, Typst documentation.
AI Social Media Dashboard	React Native · Flask · PostgreSQL · Docker · JWT	Cross-platform app with ML content recognition, real-time data visualisation, JWT auth, CI/CD via GitHub Actions, Swagger docs.
Garbage Sorting ML Model	Python · Scikit-learn · Power BI · Excel VBA	ML classifier for waste sorting. Full data-cleaning pipeline, statistical modelling.

EDUCATION

MSc Software Engineering | Aalborg University, Copenhagen Campus *2024 – 2026*

BSc Software Development | Aalborg University, Copenhagen Campus *2020 – 2023*

General Upper Secondary Certificate (STX) | Marie Kruses Skole *2015 – 2018*

EXTRA-CURRICULAR & INTERESTS

Member of IDA (Danish Society of Engineers) · Former Board Member, Political Youth Organisation (2014–2017)) · Handball Coach (2014–2017) · Competitive football & handball player